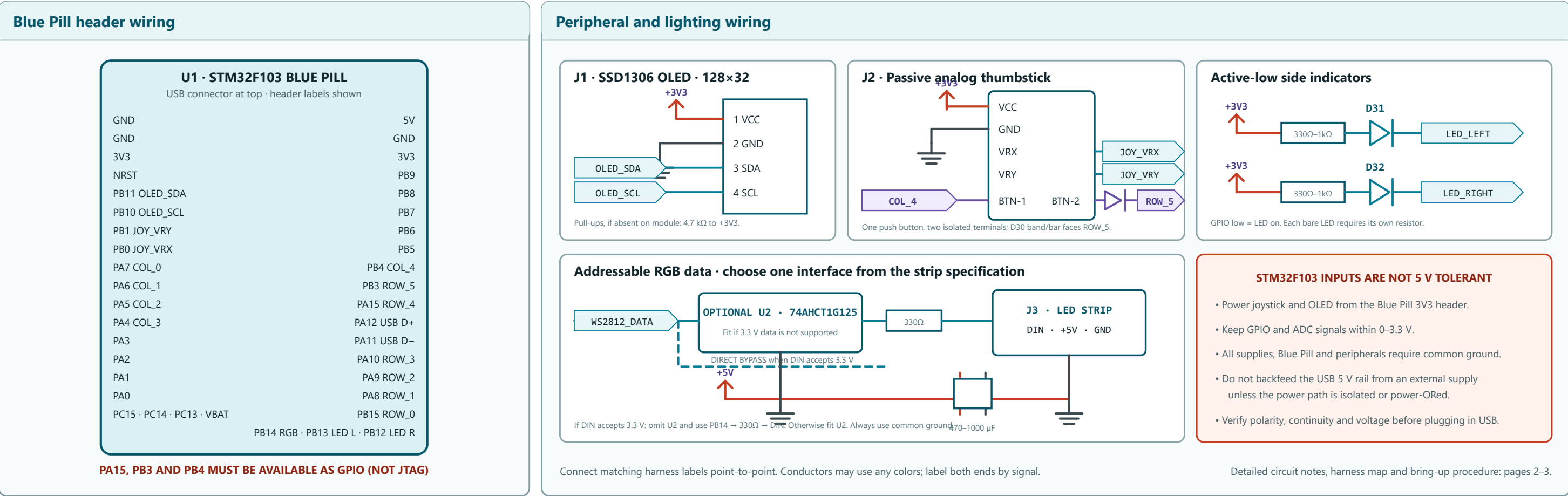
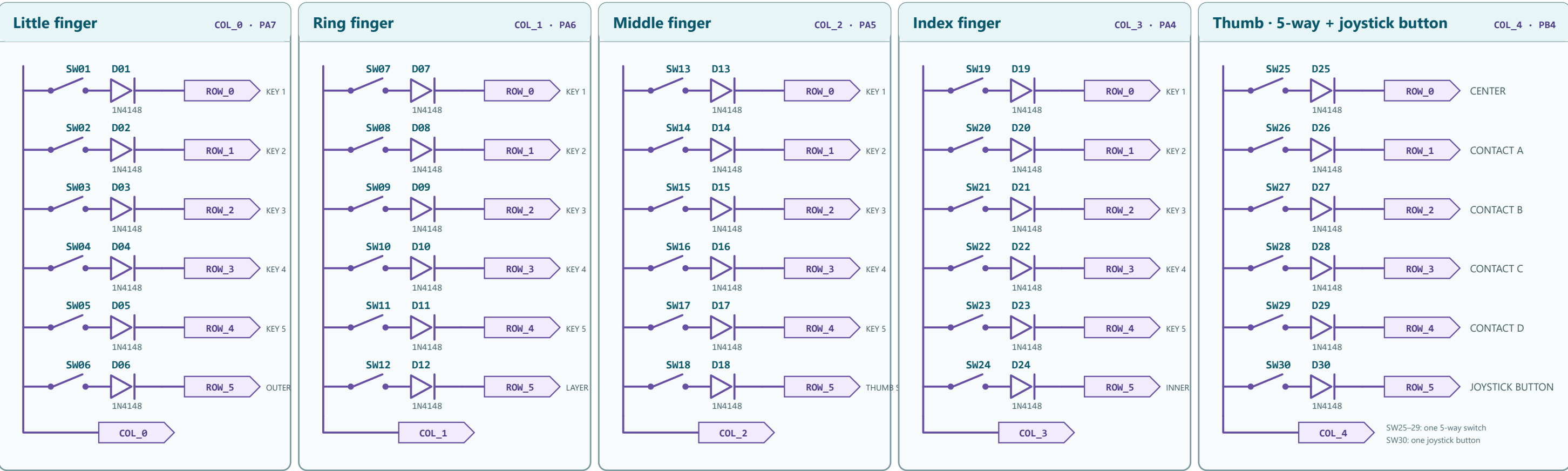
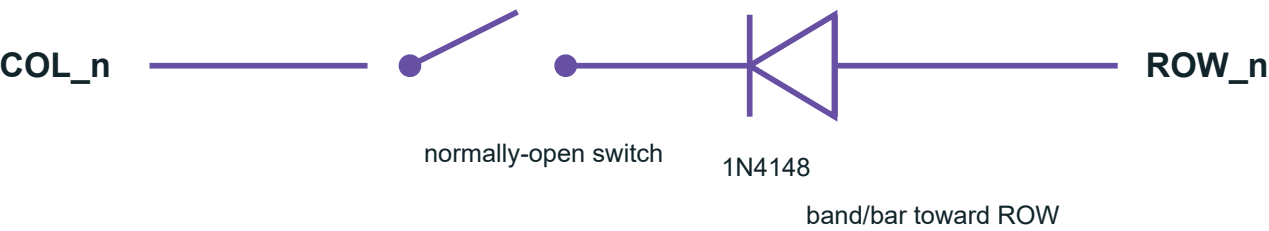




Peripheral circuits and power

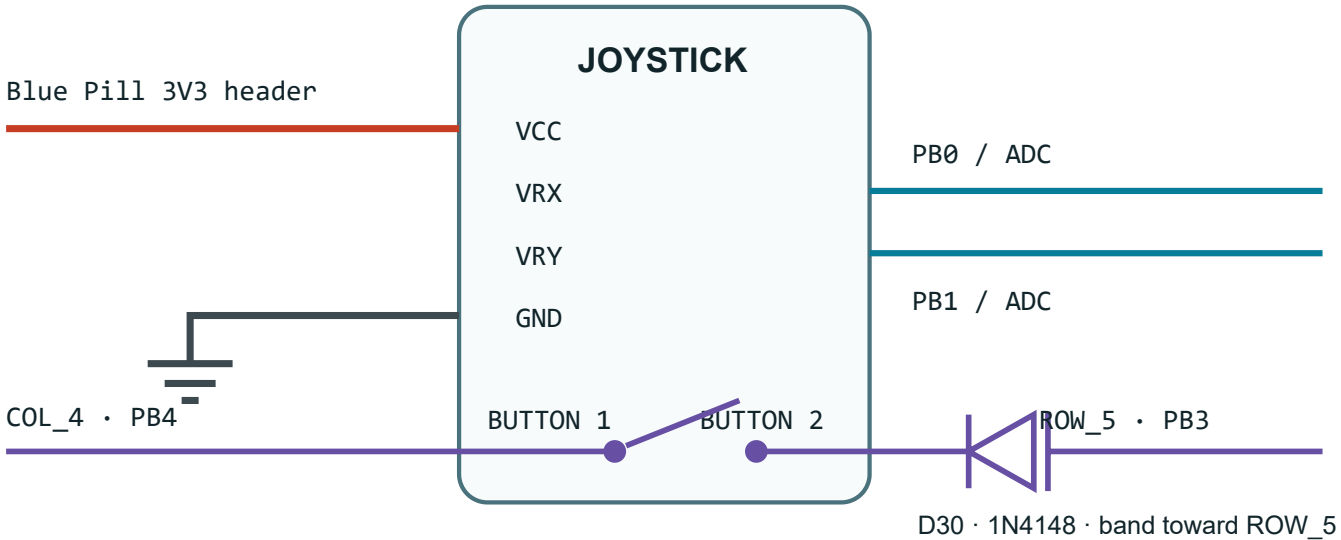
Recommended implementation details for a reliable, serviceable build

Matrix switch cell — repeat for every populated key



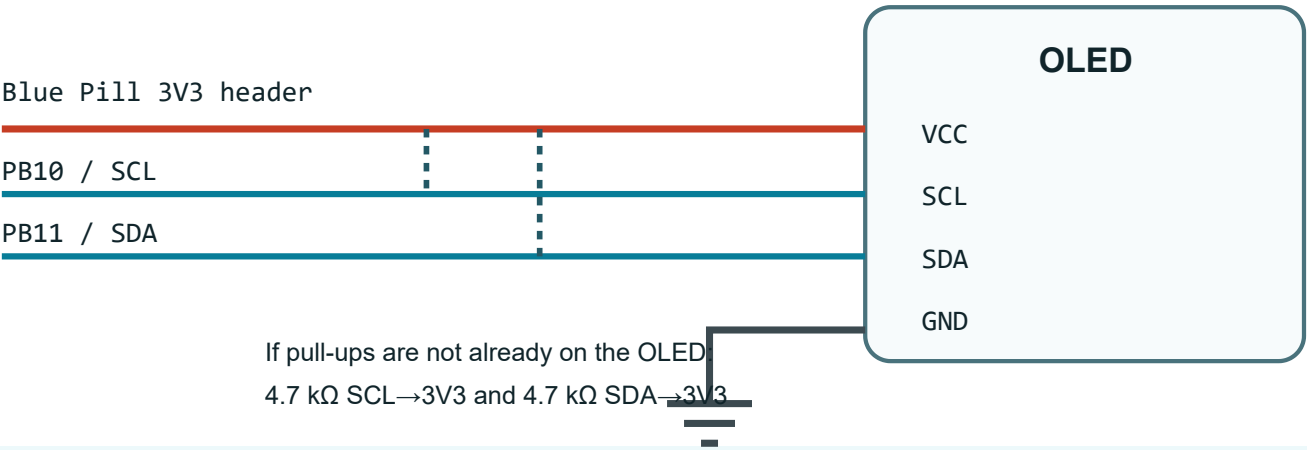
COL bus – switch – diode anode → diode cathode/bar – ROW bus
Do not share one diode between switches. All diodes must face the same way.

Analog joystick — 3.3 V only



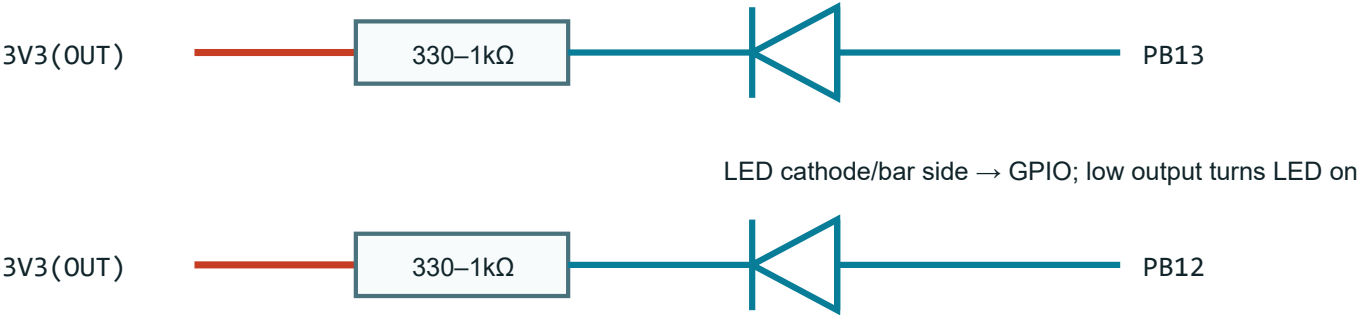
Never connect the joystick VCC wire to 5 V when VRX/VRY go directly to PB0/PB1. Measure both axis voltages first: they must stay between 0 and 3.3 V over full travel. There is one joystick push button with two terminals. Those terminals must be isolated; do not connect a module's GND-referenced SW output directly as a matrix key.

OLED I²C

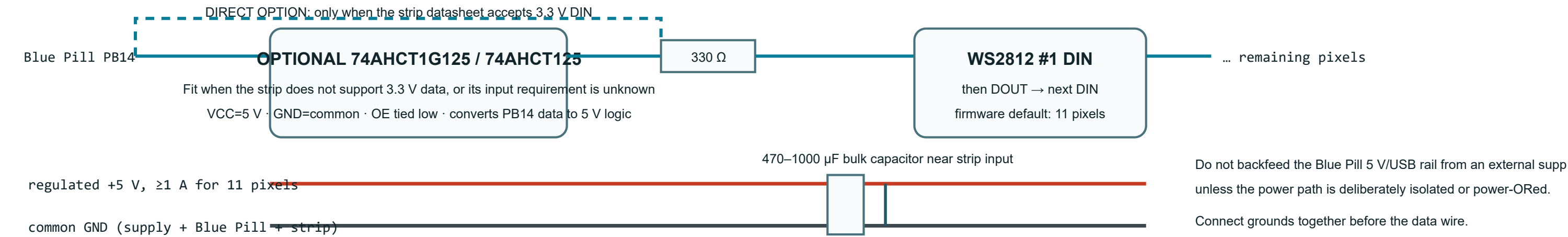


Power the module from the Blue Pill 3.3 V header unless its exact schematic proves that 5 V VCC cannot pull SDA/SCL above 3.3 V.

Side indicators — active low, matching 9R



WS2812 strip — choose the data interface from the strip specification



Harness map and bring-up checklist

Use continuity mode with all power disconnected. Choose any conductor colors, then label and verify every signal.

Matrix harness

Signal	Blue Pill GPIO	Header label
COL_0	PA7	PA7
COL_1	PA6	PA6
COL_2	PA5	PA5
COL_3	PA4	PA4
COL_4	PB4	PB4
ROW_0	PB15	PB15
ROW_1	PA8	PA8
ROW_2	PA9	PA9
ROW_3	PA10	PA10
ROW_4	PA15	PA15
ROW_5	PB3	PB3

Peripherals

Function	Blue Pill	Header label
Left side LED	PB13	PB13
Right side LED	PB12	PB12
WS2812 data	PB14	PB14
OLED SDA	PB11	PB11
OLED SCL	PB10	PB10
Joystick VRX	PB0 / ADC8	PB0
Joystick VRY	PB1 / ADC9	PB1
Joystick ground	GND	GND
Joystick click	COL_4 + ROW_5	PB4 + PB3
3.3 V peripherals	3V3	3V3
Logic ground	GND	Either GND header

Unused GPIO remains available only where it does not conflict with USB, SWD, the bootloader, or the documented JTAG-remapped matrix pins.

Reported 5-way wiring

Contact	Matrix connection
Common	COL_4 / PB4
Center	ROW_0 / PB15
A	ROW_1 / PA8
B	ROW_2 / PA9
C	ROW_3 / PA10
D	ROW_4 / PA15

Direction mapping to verify: 9R assigns the five contacts to ROW_0–ROW_4 but does not label their physical directions. The supplied harness calls ROW_0 Center, while current OLED navigation calls (0,4) Right and (1,4) Select. Identify A–D with a meter, then align firmware or wiring.

Power-off checks

1. Disconnect USB and every external supply.
2. Check for a short between 3V3 and GND, then 5V and GND.
3. Verify the BOOT0 jumper is in its normal 0 position. BOOT0 and BOOT1/PB2 are not part of the key matrix.
4. Continuity-check every harness wire to the correct Blue Pill GPIO header label.
5. Verify every matrix diode band faces the ROW bus.
6. Verify PA7 and PB15 are not permanently shorted together.

Safe staged bring-up

1. Disconnect OLED, joystick, WS2812 strip, and side LEDs. Connect only Blue Pill USB.
2. Enter STM32duino DFU, flash the STM32 .bin, then reset or reconnect as required and confirm the USB HID device enumerates.
3. Add the matrix harness. Test one switch at a time in Vial/WebHID.
4. Add joystick powered from 3V3. Measure VRX/VRY before connecting PB0/PB1.
5. Add OLED at 3.3 V and confirm SDA/SCL idle near 3.3 V.
6. Add side LEDs with resistors.
7. Add the externally powered WS2812 strip last. Connect PB14 directly through 330 Ω only if the strip supports 3.3 V data; otherwise fit the level shifter.

Design basis: based on the validated 9R STM32 wiring and the maintained firmware pin map. It preserves the COL2ROW topology, COL_4 thumb common, ROW_5 stick click, 3.3 V analog/OLED supplies, active-low indicators, and 5 V WS2812 supply. The level shifter is optional when the exact strip supports a 3.3 V data input; otherwise it is required for dependable level conversion. The data resistor, bulk capacitor, and separated LED-power guidance are reliability additions. This is not a certification drawing; confirm actual module datasheets.